Manifold for T1D – Restoration

β-cell preservation, regeneration, and replacement-therapy analytics

T1D does not end at diagnosis — three clinical paradigms now converge on restoring β-cell function: **preservation** (teplizumab post-diagnosis, verapamil, ATG, GLP-1 combos), **regeneration** (DYRK1A inhibitors, senolytics), and **replacement** (Vertex VX-880, Sernova Cell Pouch, ViaCyte/CRISPR). Each has actionable agents and rich longitudinal data; the analytic gap is matching patient to therapy and ordering combination regimens on a principled causal graph. **Manifold** configures three engine capabilities to the domain — causal discovery on multi-omic + trial data, criticality estimators on post-intervention trajectories (mixed-effects C-peptide, heterogeneous-treatment-effect learners, GP regression, change-point), and Pearl do-calculus / interdiction for rational combo design.

> Vertex VX-880's first patients achieving insulin-independence is a proof-of-principle, not a finished product. The question is no longer "can β-cell function be restored?" — it is "for whom, with what adjuncts, and for how long?" That question is causal-inference-shaped.

ΩF PILLAR MAPPING – RESTORATION PHYSIOLOGY

ΩF PILLAR	RESTORATION READING
I — Interaction	Drug × genotype × autoantibody-profile interactions for combo response; graft × host-immune compatibility
R — Reserve	Residual / post-infusion β-cell mass (C-peptide AUC on MMTT); stress-test response
J — Junction	Immune–metabolic coupling: cytokine-panel ↔ C-peptide transfer entropy; autoimmunity × graft survival
C — Cascade	β-cell senescence and ER-stress cascades; DYRK1A → cell-cycle progression; inflammatory escalation post-graft
T — Temporal	Hazard of honeymoon-end, graft-failure, or response-loss given current trajectory

THE ASK

Partnership with a preservation-trial, β-cell-regeneration, or islet-transplant PI (or cohort steward — Joslin, Mount Sinai Stewart Lab, Vertex trial collaborator, HIRN consortium member) for:

1. Retrospective trial or registry data access under their existing Data Use Agreement.
2. Clinical co-authorship on the HTE / graft-survival benchmark paper (12–14 weeks from data access).
3. Co-application to Breakthrough T1D Cures, Helmsley Restoring Insulin Production, or NIH NIDDK HIRN for the v2 prospective arm.

Apex Analytica brings the causal-inference engine, the **Ω**F framework, and the software team. Full brief: `M-T1D-02`.